

DEVELOPMENT OF THE TARANAKI BASIN AND COMPARISONS WITH THE GIPPSLAND BASIN: IMPLICATIONS FOR DEEPWATER EXPLORATION

C. Uruski¹, P. Baillie² and V. Stagpoole¹

¹Institute of Geological and Nuclear Sciences Ltd
PO Box 30-368

Lower Hutt New Zealand

²TGS-NOPEC Geophysical Company

Level 5

MLC House

1100 Hay Street

West Perth WA 6005

c.uruski@gns.cri.nz

peterb@tgsnopec.com.au

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INTRODUCTION

New Zealand's Taranaki Basin is named after Mount Taranaki, a large Pleistocene volcano forming the core of the Taranaki Peninsula and surrounded by oil and gas fields (King and Thrasher, 1996). The basin lies partly onshore but mostly offshore below the broad continental shelf to the west of central North Island. Taranaki, New Zealand's only petroleum producing basin so far, is separated from the Northland Basin (Isaac et al, 1994) that occupies the Northland Peninsula and its relatively narrower shelf, only by an arbitrary east-west line at about 37°S (Fig. 1). The Astrolabe survey, named after the ship commanded by an early French explorer of New Zealand waters, Dumont D'Urville, confirmed that the Taranaki Basin extended into the head of the New Caledonia Basin as previously suspected (Uruski and Wood, 1991; Uruski, 2000). The Taranaki and Northland Basins, together with the deepwater extension of the Taranaki Basin, are essentially one large basin system, referred to in this paper as the greater Taranaki Basin.

On the Australian side of the Tasman Sea (Fig. 1), the Gippsland Basin, has its onshore, shelf and deepwater portions that have long been considered part of the same depositional system. Gippsland is also Australia's most productive basin with total reserves, including production to date, of about 10 trillion (10^{12}) cubic feet (TCF) of gas and four billion (10^9) barrels (BBL) of oil. Comparable figures for Taranaki are 7 TCF of gas and 500 million BBL of oil.

With locally important petroleum basins on either side of the Tasman Sea, many authors have sought to compare the two (e.g. Sutherland et al, 2001), as valid analogies can clarify poorly-understood factors. Before the Astrolabe survey, these comparisons could only be partial and analogies were inconclusive.

The new survey revealed that the thick sediments lying below the top Cretaceous reflector are laterally extensive, filling a depocentre about 100 km across and at least 150 km along the axis of the basin. Seismic evidence is presented for a Strzelecki Group equivalent as well as a prograding succession similar to the Emperor and Golden Beach sub-Groups.

ABSTRACT

Exploration of the Taranaki Basin entered a new phase in 2001 with Astrolabe, a 6,200 km high-quality 2D seismic survey acquired by TGS-NOPEC that has outlined a large depocentre containing up to 10 km of sedimentary fill. This new data has extended the previously-known Taranaki Basin into deeper water beyond the shelf edge. Subsequently, the New Zealand Government released an area of 42,000 km² for competitive bidding to close in September 2003.

Sequence analysis shows that a major deltaic system, comparable to the Golden Beach and Emperor subgroups of the Gippsland Basin, built into a restricted seaway during the Late Cretaceous and culminated with deposition of the Rakopi Formation coal measure succession. The Rakopi Formation covers an area of at least 15,000 km² of the study area and was followed by a transgression that continued until the Miocene.

Minor Eocene folding created broad structures with potential to trap large volumes of petroleum. Other potential trapping structures include drape across Cretaceous rift blocks and turbidite mounds of Miocene age.

Modelling shows that much of the Early Cretaceous delta is thermally mature and should be expelling petroleum today. Reservoir facies are present at many horizons, but the primary target is expected to be sandstones of the Rakopi Formation coal measures.

Many analogies can be drawn between the Taranaki and Gippsland basins. The deepwater Taranaki basin appears to be equivalent, however, to the offshore, oil-prone part of Gippsland while the nearshore Taranaki and Great South basins together form an analogy for the more gas-prone nearshore part of Gippsland.

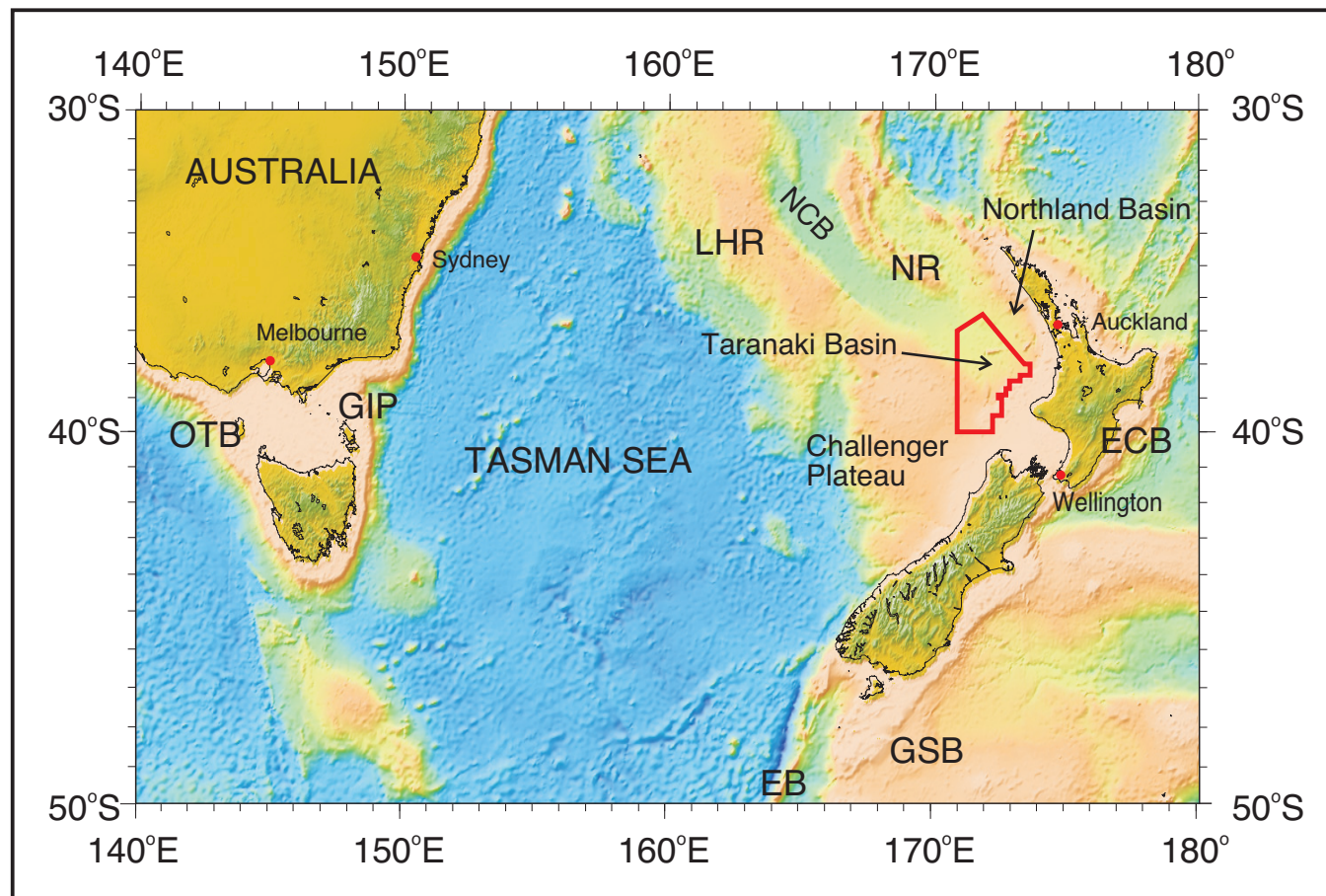


Figure 1. Map showing locations of major features discussed. The red polygon is the area of the Astrolabe survey. OTB—Otway Basin, GIP—Gippsland Basin, LHR—Lord Howe Rise, EB—Emerald Basin, NCB—New Caledonia Basin, NR—Norfolk Ridge, GSB—Great South Basin, ECB—East Coast Basin. After Stagpoole (2002) and Smith and Sandwell (1997).

Background to the Astrolabe survey

A single-channel reconnaissance survey in the head of the New Caledonia Basin suggested to the authors (Uruski and Wood, 1991) that the Taranaki Basin extended beyond the shelf edge, the previous limit for petroleum exploration. This was confirmed when, in 1996, the New Zealand Government acquired a single 1,000 km seismic line along the axis of the New Caledonia Basin from the Wainui-1 well on the Taranaki shelf edge as part of a joint project with AGSO (now Geoscience Australia). The deepwater Taranaki sector of this line was interpreted to be underlain by Cretaceous sediments, up to 5,000 m thick (Uruski, 2000).

In 2001 TGS-NOPEC Geophysical Company, in collaboration with the Institute of Geological and Nuclear Sciences (GNS) acquired 6,200 km of new non-exclusive 2D data in the deepwater Taranaki Basin (Fig. 2). The survey was tied to eight wells on the Taranaki shelf and the deepwater Wakanui-1 well in the adjacent Northland Basin.

TGS-NOPEC and GNS planned the Astrolabe 2D reconnaissance survey of the deepwater Taranaki Basin on a basic 10 km grid. In water depths beyond 1,500 m and in the southern sector of the area the grid is much wider. The survey was acquired to confirm the existence of a deepwater extension to Taranaki Basin and to investigate its petroleum prospectivity.

Aims of this paper

First, we investigate comparisons of the Gippsland and Taranaki Basins possible prior to acquisition of the new Astrolabe dataset by considering plate tectonic and tectonostratigraphic evolution of the two basins. We then show how the new data can improve the analogy. Finally, implications for petroleum prospectivity of the two basins are discussed.

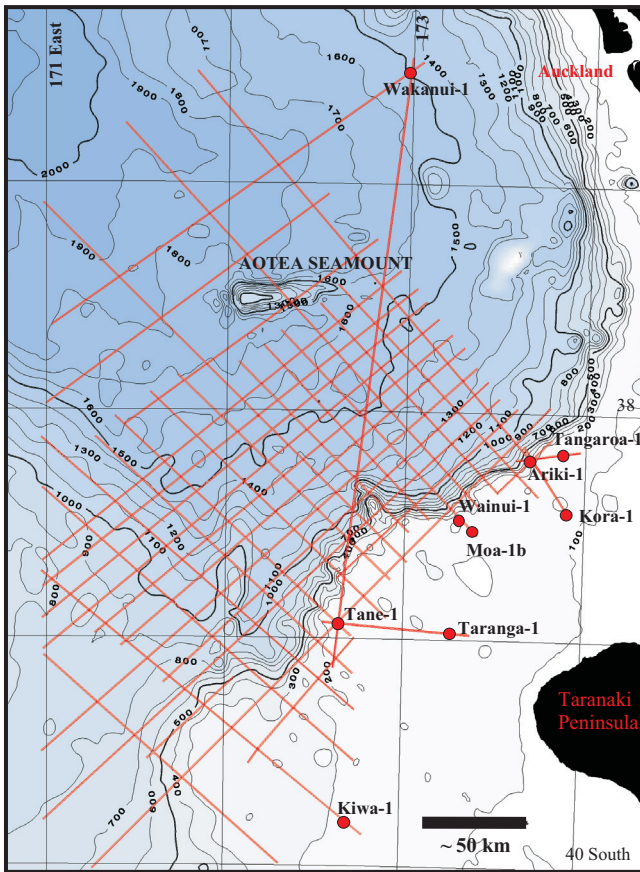


Figure 2. Map showing Astrolabe survey lines with relevant wells and regional bathymetry. The large seamount centred near 172.25°E 37.5°S is the Aotea Seamount, a large Miocene volcano.

PRE-ASTROLABE COMPARISONS

Plate tectonics

Plate tectonics provided a basic control on basin formation across large regions. If the same event initiated both the Gippsland and Taranaki Basins, they are by definition, generically related. Although the New Zealand mini-continent formed part of the Gondwana margin until the Late Cretaceous when it was rifted from Australia and Antarctica (Fig. 3), the known tectonic and sedimentary history of its basins was thought to have started later than in Gippsland.

Tasman seafloor spreading started in the Late Cretaceous (85 Ma) and finished by the end of the Paleocene (~50 Ma). The New Zealand region then foundered until the modern Pacific/Australia plate boundary was initiated near the start of the Miocene. A complex plate margin then formed with the Pacific Plate being subducted below North Island and the Australian plate below the southern end of South Island. Between the two subduction zones runs the Alpine Fault, a major

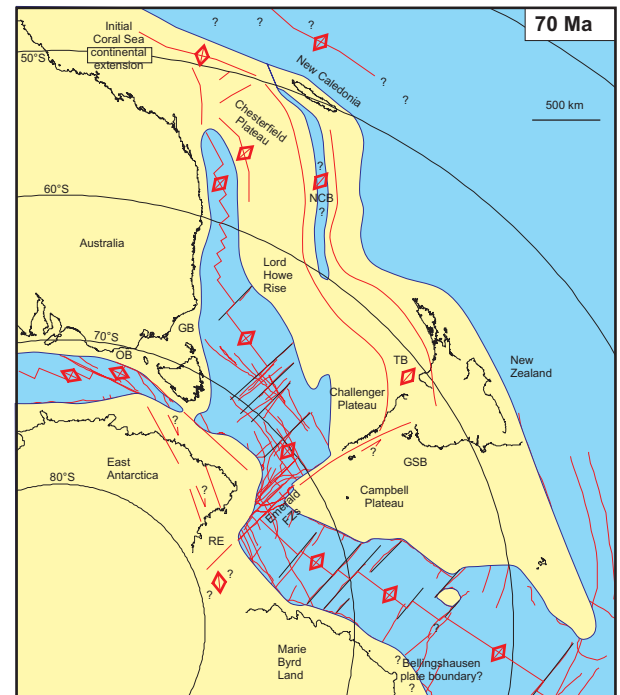
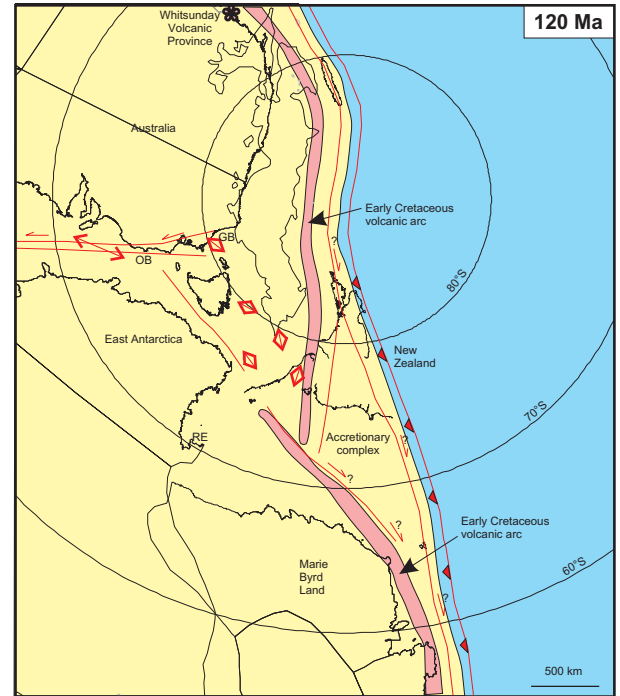


Figure 3. Plate reconstructions for 120 Ma and 70 Ma (after Sutherland et al., 2001). GB Gippsland Basin, TB Taranaki Basin, GSB Great South Basin

strike-slip plate boundary, along which the southern Alps have been uplifted some 30 km to provide a voluminous source of clastic sediments to New Zealand's sedimentary basins.

Three basin-forming rift phases are recognised in the region; the Early Otway, Tasman and Taranaki (Sutherland et al, 2001). The Early Otway Phase lasted from 160–100 Ma. This Late Jurassic and Early Cretaceous phase was concurrent with convergence and arc volcanism along the eastern margin of Gondwana. Volcanism continued in the New Zealand region until about 130 Ma (Mortimer et al, 1999). Southern Australia started rifting away from Antarctica during the Late Jurassic and this phase is recorded by initial basin formation in the Gippsland Basin and deposition of the Jurassic Beds and the Strzelecki Group. Before the Astrolabe survey, there was little evidence of extensional tectonics in New

Zealand at this time and sedimentary rocks of this age were unknown in the Taranaki Basin.

The Tasman Rift Phase was essentially a Late Cretaceous event lasting from latest Albian–Early Campanian time (100–80 Ma). This phase was recorded by a change in regional tectonics led up to the spreading of the Tasman Sea (Norvick et al, 2001). In the Gippsland Basin, thick fault-bounded units of clastic sediments, the Emperor and Golden Beach Groups, were deposited. Several comparable New Zealand units were present. The Hoiho Group (Fig 4) was deposited in the Great South Basin, while the Taniwha Formation (Shell, BP and Todd, 1986) was deposited in Taranaki and Northland. Samples of coal measure rocks of late Cenomanian to Coniacian age (95–86.5 Ma) were dredged from the flank of the West Norfolk Ridge (Herzer et al, 1999).

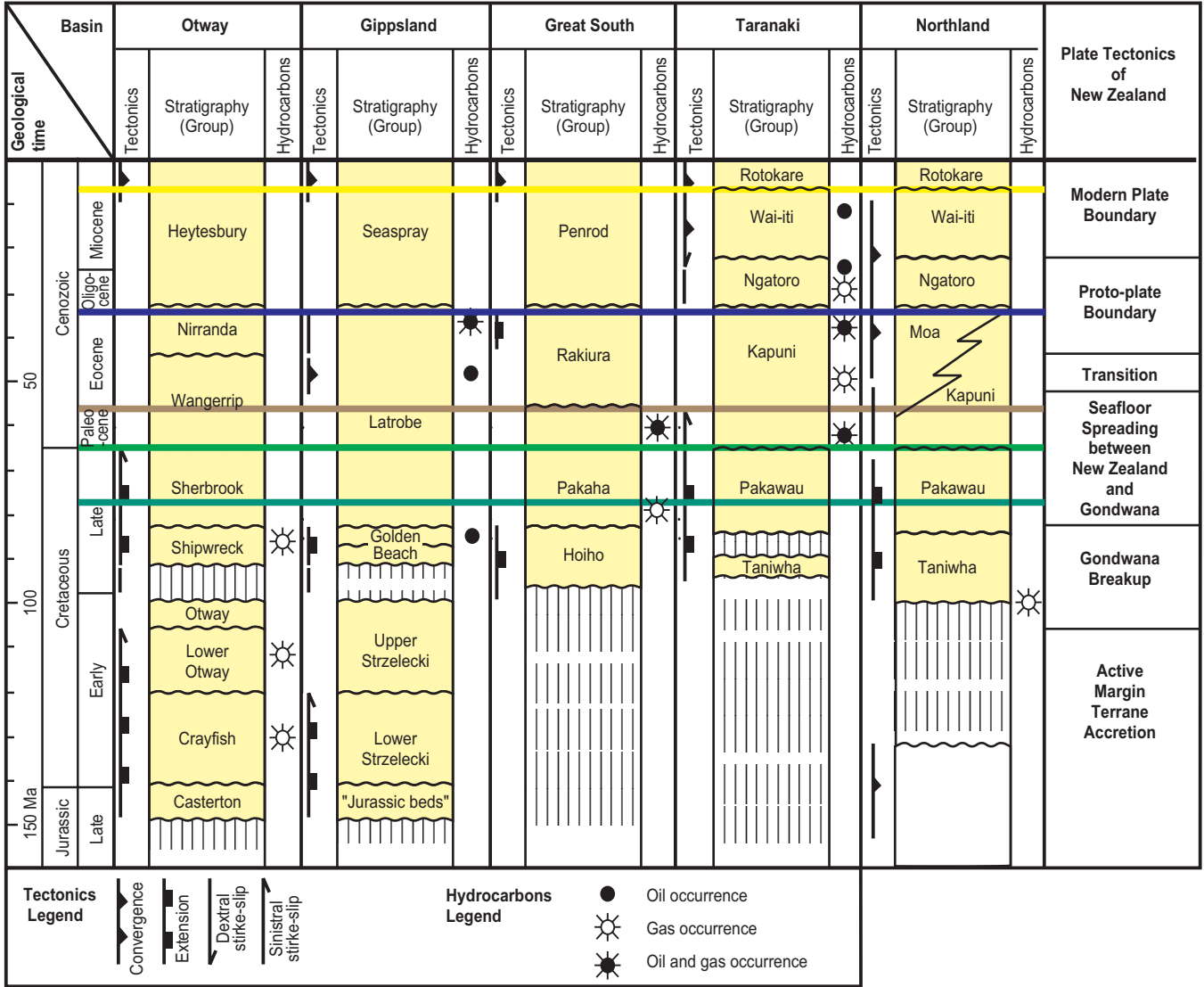


Figure 4. Comparison of stratigraphy of Southeast Australian and New Zealand basins. This does not include the new information from the Astrolabe survey.

The Taranaki Rift Phase continued for the rest of the Late Cretaceous until mid-way through the Paleocene despite concurrent seafloor spreading in the Tasman Sea. Extension was widespread in New Zealand through this time, particularly in Western Southland (Turnbull et al, 1993), probably in response to the opening of the Emerald Basin to the south of New Zealand. This phase does not appear to have affected the Gippsland Basin, which was separated from the New Zealand mini-continent by the Tasman Sea.

Although structural inversion is recorded by extensive erosional channels in a pre-existing seafloor near the start of the Eocene (Johnstone et al, 2001), little tectonic activity is apparent in Gippsland later than the middle of the Late Cretaceous. In contrast, the present-day plate margin is almost entirely responsible for the presence of New Zealand, and much of the evolution of New Zealand and its sedimentary basins took place during the Neogene. Many spectacular compressional structures were formed at a variety of scales from petroleum trap-forming folds in sedimentary basins to uplift of the Southern Alps, which resulted in the influx of large volumes of clastic sediments while offshore Gippsland was being starved of clastics.

Stratigraphy

Basement in New Zealand is arranged into a series of linear terranes about parallel with the ancient Gondwana margin. Consequently, basement composition in Taranaki is variable. In the north around the Ariki-1 well (Shell, BP and Todd, 1984) basement consists of metamorphosed igneous and volcanoclastic sediments of Mesozoic ages, correlated with the Median Tectonic Zone (Mortimer et al, 1997) and the New England Fold Belt of eastern Australia. The Wainui-1 well stopped in metamorphosed siliceous sandstone of the Paleozoic Takaka terrane, while further south in Tane-1 and Kiwa-1 basement is of Early Cretaceous Separation Point granite.

The oldest sediments in Gippsland Basin (Norvick et al, 2001) are the coal measures and lacustrine facies of the Jurassic Beds, deposited in extensional basins during the early stages of the Early Otway Rift Phase. This rift phase continued through the Early Cretaceous when the Strzelecki Group was deposited. Gippsland experienced a mainly Campanian (100–90 Ma) unconformity followed by deposition of the Emperor and Golden Bay subgroups in the Late Cretaceous.

The oldest known unmetamorphosed sedimentary rocks below the Taranaki shelf are the Cretaceous Taniwha Formation coal measures (SBPT, 1986) deposited in extensional basins in variable basement rocks. The Taniwha Formation is of Latest Albian to Turonian age (100–90 Ma). This sequence is considerably older than the Pakawau Group, which forms the bulk of Cretaceous deposits below the Taranaki shelf, is of Late Campanian age (80–65 Ma). The Pakawau Group of the Taranaki Shelf consists of the Rakopi and North Cape formations. The former is a coal measure succession, while the North Cape Formation is generally of marine rocks.

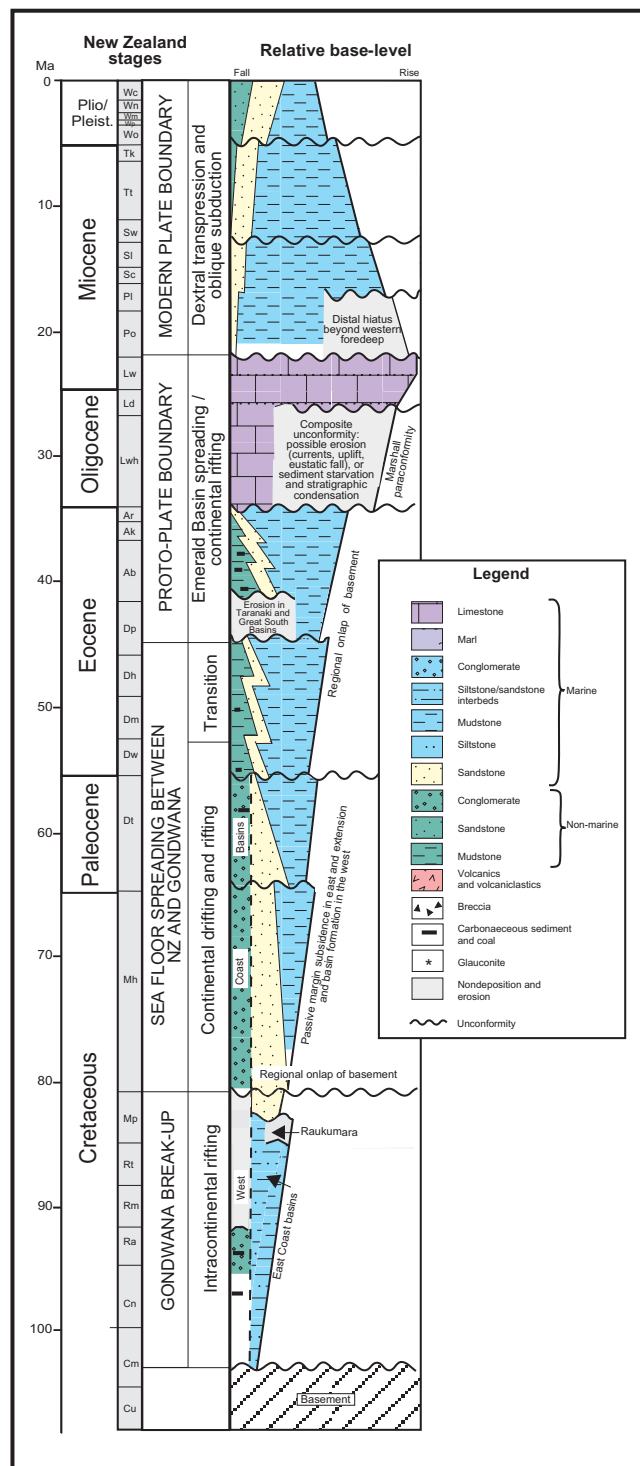


Figure 5. Generalised stratigraphy of New Zealand Basins as known before the Astrolabe survey. This figure shows the overall transgression that continued through the Cretaceous and culminated in the Oligocene, followed by regression resulting from development of the Neogene plate boundary.

Syn-rift megasequence A

The basal unit is clearly a syn-rift megasequence as it fills grabens within basement and overtops some horst blocks (Fig. 6). The top of this unit is marked by downlap of the overlying megasequence and it does not exist landward of the present shelf edge. Its internal reflectors are discontinuous and of variable amplitude. A large volcanic edifice was extruded onto its top surface. The age of the syn-rift megasequence is poorly constrained. The oldest sedimentary rocks encountered in nearby wells are the Rakopi Formation of Campanian age (about 75 Ma, Fig. 7), but the syn-rift megasequence is clearly much older. A search for possible analogous rocks yields the Taniwha Formation of Taranaki and Northland and the Late Jurassic coal measures of the Huriwai Group (Ballance, 1988) of the Murihiku Group that were previously considered as part of the basement terrane. The Huriwai Formation may be the basal unit of megasequence A. The Wakanui-1 well drilled by Conoco in 1998 reputedly penetrated Jurassic coal measures above basement. If true, this is strong evidence for Jurassic sediments being present in the deepwater part of the greater Taranaki basin. A Gippsland equivalent for the major part of megasequence A is likely to be of Early Cretaceous making the Strzelecki Group a prime candidate.

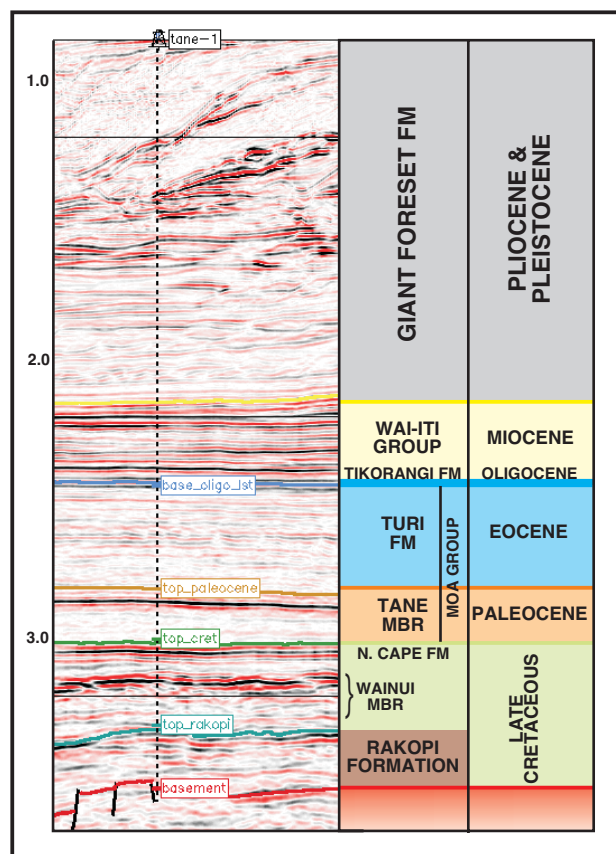


Figure 7. Seismic line tie to Tane-1 showing main sedimentary units occupying the shelf area of Taranaki basin.

Prograding megasequence B

The top of the prograding megasequence is tied to the Rakopi Formation in Tane-1 (Fig. 7), which dates the culmination of this prograding succession as Late Campanian (about 75 Ma). The prograding megasequence is subdivided into five bodies (Fig. 6) starting with a high-stand systems tract. This unit is partially fault-bounded below the present shelf edge zone and thickens to nearly one second twt before tapering to zero near the basin centre. This basal unit may include much coarse clastic material shed from the fault scarps.

The second unit is a more typical slug-model, high-stand systems tract, thickening from zero on top of unit one to about 750 milliseconds twt above the toe of unit one, before thinning to zero again some 50 km to the northwest. Internally, sinusoidal reflectors onlap the top surface of the older systems tract and downlap to the northwest. Together, the two older units are symptomatic of continuing, if episodic, relative sea-level rise, probably tectonically driven by extensional of the Tasman Rift phase 100–80 Ma (King et al, 1999; Norvick et al, 2001).

The third unit is a low-stand wedge with complex internal reflectors suggesting good development of seabed fan turbidites. The unit thickens to the northwest from zero at the slope break of unit two to nearly 500 milliseconds twt. Reflectors also onlap the older volcanic mound.

Unit four is another high-stand systems tract exhibiting prograding internal reflectors. The topset beds in this unit are prominent in that they are of high amplitude, discontinuous reflectors similar to coal measure units elsewhere in Taranaki. Further, the top surface of this unit ties to the Rakopi Formation coal measures in Tane-1.

The final unit that lies below the top Rakopi Formation reflector is a thin low-stand wedge located to the northwest of unit four and onlapping its toe.

Together, the prograding reflector geometry and time isopachs (Fig. 8) of megasequence B, which clearly illustrates a lobate compound body, are compelling evidence for deltaic deposition. The evidence suggests that a river or rivers fed into a restricted seaway at the head of the New Caledonia Basin. Such seismically-imaged deltas have been notably absent from the mid-Cretaceous basins of New Zealand, despite abundant evidence of fluvial deposition. Much of the Hoiho Group (Fig. 4) of the Great South Basin, for example, is fluvial, but no outlet to the sea is apparent. The Great South Basin was adjacent to Taranaki during this period so the Taranaki delta may be the outlet for the Hoiho Group rivers.

North Cape Formation—megasequence C

Until the Rakopi unit was deposited, sediment supply was able to keep up with relative sea-level rise. By the Late Campanian, the balance had been disrupted by either an acceleration of relative sea-level rise or a

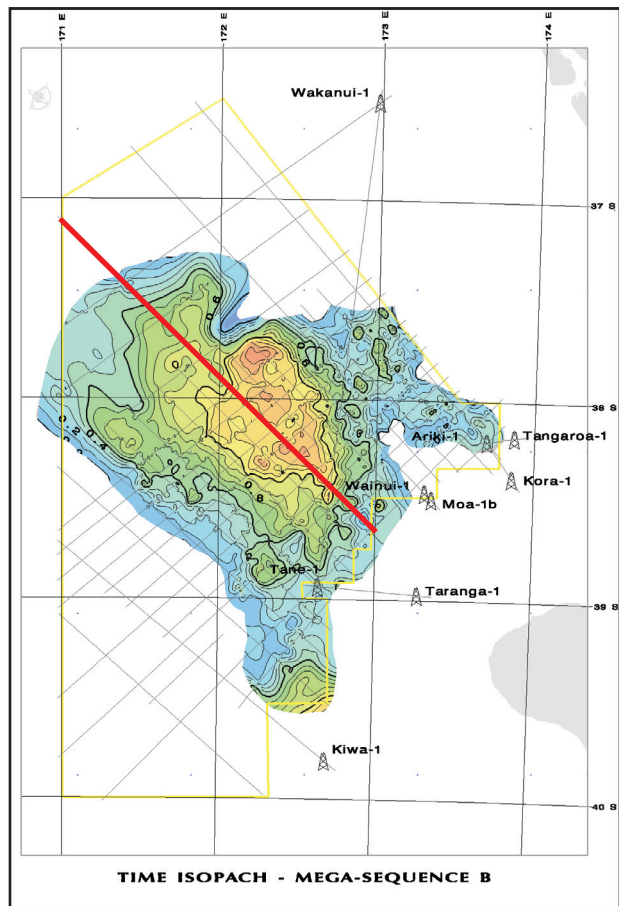


Figure 8. Time isopach map of mega-sequence B, the Late Cretaceous delta. The red line is the line of figures 6 and 8.

reduction in sediment supply. The general transgression noted in the Taranaki Basin and elsewhere was under-way and the locus of deposition shifted landward by about 100 km, resulting in deposition of the mainly marine North Cape Formation. Previously, an extensive continental shelf and coastal plain was established, so with this relatively flat topography, minor changes of sea-level and sediment supply resulted in large landward and seaward shifts in depositional systems.

Paleogene

The oscillations of depositional systems continued throughout the rest of the Late Cretaceous and Paleogene. During this time, the surrounding basement highs of the Challenger Plateau and part of the West Norfolk Ridge Basin as well as the more remote New Zealand mainland, appear to have been emergent and supplying sediments to the basin. Folding of Eocene age formed large, gentle anticlines and flower structures as older faults were reactivated as compressional and strike-slip faults. By the Oligocene, clastic sediments were in very short supply and carbonate deposition was widespread.

Neogene

Wells at the shelf edge show that carbonate deposition was not unknown during the Miocene. The deepwater Taranaki area remained in relatively deep water throughout the Neogene. Miocene depocentres reflect turbidite deposition, particularly across the paleo shelf edge to the northwest. Despite the renewed sediment supply due to activity at the plate boundary, most Miocene sediments were deposited closer to shore. The influx of sediments associated with renewed Alpine uplift in the Pliocene formed the present shelf. The Pliocene within the study area is up to one second twt thick and is mainly clastic, consisting of a high proportion of seafloor fan turbidites.

Petroleum potential of deepwater Taranaki Basin

In deepwater Taranaki, thick coal measure facies at top Rakopi level cover an area of at least 15,000 km². Extensive coals are also likely within the topset beds of deeper prograding units of the Late Cretaceous delta, while disseminated coaly kerogen is likely to be present throughout the foresets and to have been carried into deeper water by low-stand turbidite fans. The delta was built into a restricted seaway between the emergent Challenger Plateau and the West Norfolk Ridge suggesting that Cretaceous marine mudstones are also possible petroleum source rocks. If equivalents of the Jurassic Huriwai Formation are present in the basin, a further Jurassic source is also likely along the flanks of the basin, where it has not been buried too deeply.

Potential petroleum reservoirs in deepwater Taranaki include the sandstones of the Rakopi coal measures themselves and marine sandstones of Late Cretaceous North Cape Formation equivalents as well as Cretaceous and Paleogene turbidite fans.

Thermal modelling shows that source rocks within the delta are generally mature today and that maturity is a low risk for this basin. Two-dimensional thermal maturity modelling predicts there is up to 1,000 m of mid-Cretaceous source rock in the oil expulsion window (Fig. 9). The present day oil generation window is below about 2,800 m below sea-floor (bsf) for marine source rocks (Type II kinetics) and 3,200 m for non-marine source rocks (Type III kinetics). The present day gas window is below about 5,000 m bsf for non-marine source rocks (Type III kinetics).

In the centre of the basin, oil expulsion started in the Cretaceous and Paleogene. A post-Miocene pulse of sedimentation resulted in the top of the oil expulsion window climbing a further 200 m to reach much of the Rakopi Formation equivalent at the present day. The gentle Eocene tectonism formed trapping structures as petroleum was being expelled from source rocks and the negligible Neogene tectonism that affected the deepwater Taranaki Basin makes it likely that trap integrity would have been maintained.

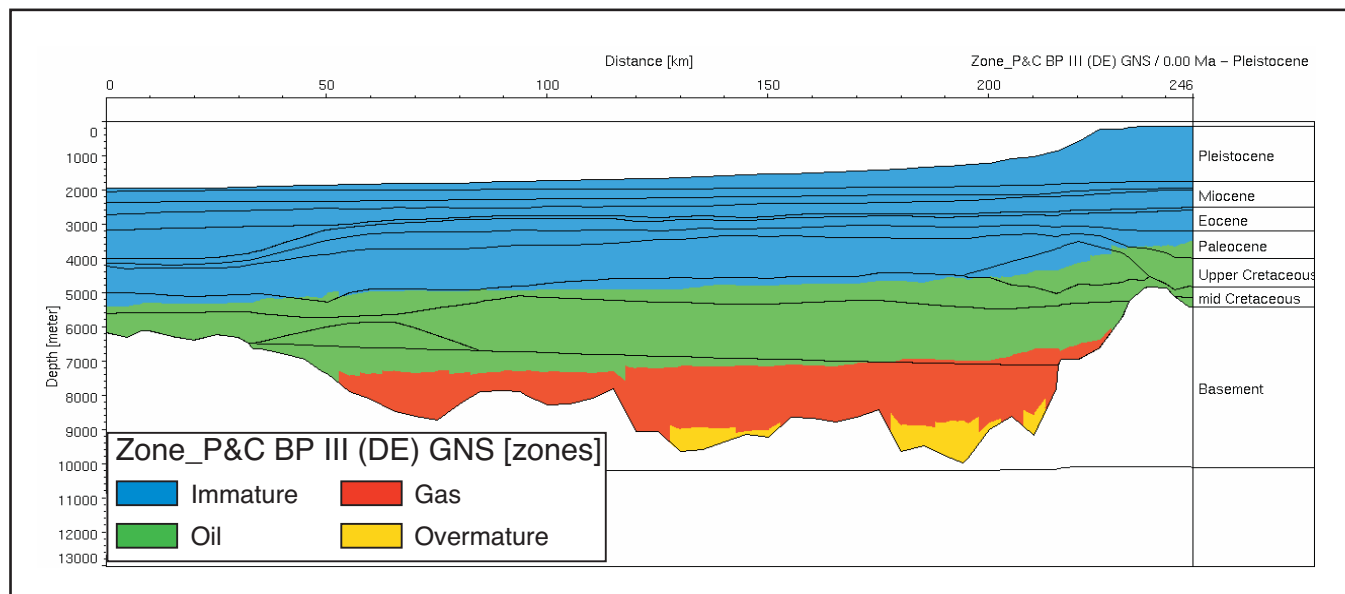


Figure 9. Present day maturity model based on Pepper and Corvi (1995) kinetics. The model was generated using IES Petromod software.

Oil or gas?

Following the discovery of the onshore Kapuni gas and condensate field in the 1950s and the giant Maui gas and condensate field in the 1970s, the industry tended to the belief that Taranaki was gas-prone and had therefore not been a popular exploration venue. Several lines of evidence suggest that the deepwater Taranaki Basin should be oil-prone rather than gas-prone.

The first is that the shallow water and onshore Taranaki Basin is now seen as more oil-prone than previously thought. This is due to an oil trend in the onshore overthrust zone, the Kaimiro and Ngatoro oilfields in northern onshore Taranaki, the Maari discovery in 1999 and the more recent Rimu discovery in 2000. In addition, the recent offshore discoveries of an oil leg in the Maui field and a better understanding of its charge history enhance the Taranaki Basin's attraction as an oil basin.

Maui hydrocarbons have a terrestrial source, predominantly Late Cretaceous coal and coaly sediments (Sykes, 2001), and a complex charge history (Funnel et al, 2001). Earliest charge was hot brines with high CO₂ content, followed by several phases of oil migration and a latest phase of oil and gas charge.

The second line of reasoning is based on the seismic stratigraphy outlined above. The increased marine influence implied by the delta interpretation and the known oil richness of deltas around the world increases confidence that the deepwater Taranaki Basin is oil-prone. In addition, the inferred restricted seaway increases the chances of widespread organic rich shales being deposited.

POST-ASTROLABE COMPARISON WITH GIPPSLAND

Tectonostratigraphy

The Astrolabe survey provides new data to help comparisons between the greater Taranaki and Gippsland Basins. A direct comparison between the Jurassic Beds of Gippsland and the Huriwai Formation of Northland is possible in the deepwater Taranaki Basin. In both basins they appear to be essentially coal measures or lacustrine and their possible presence in Taranaki suggests that both basins originated during the Late Jurassic in response to the Early Otway Rift Phase. The analogy is strengthened by correlation of the remainder of Megasequence A with the Strzelecki Group.

The proposed Astrolabe delta of Megasequence B was deposited in Late Albian to Campanian time (100–75 Ma) and is a time correlative of the Emperor and Golden Beach sub-groups. Megasequence B reflectors downlap onto the top of Megasequence A, providing evidence for non-deposition across much of the Strzelecki Group equivalent in Taranaki. We suggest that this may be the equivalent of the Cenomanian hiatus of Gippsland Basin. The Kipper volcanics, extruded on top of the Golden Beach Group, also has its equivalent in Taranaki (Fig. 6). The Emperor and Golden Beach sub-groups are separated by a landward shift of facies belts while in Taranaki, the boundary between the Taniwha equivalent and the overlying Rakopi Formation is a low-stand followed by a marked landward shift of facies belts and renewed delta prograding. In deepwater Taranaki, tectonics appear to have been waning through Rakopi time as would be expected, following plate separation and accompanying opening of the Tasman Sea. This allows a decreasing

sediment supply from New Zealand's eroding highs to keep pace with moderate relative sea-level rise enabling a rich flora to develop across a broad coastal plain.

The Late Cretaceous part of Gippsland's Latrobe Group exhibits a seaward shift in facies belts in response to an apparent marine regression. By this time, the Taranaki Rift Phase had commenced below the Taranaki Shelf, where a mix of coal measure and marine facies were being deposited in fault-bounded lows. The Pakawau Group coal measures and North Cape Formation marine units of Taranaki are equivalents of the Late Cretaceous Latrobe Group. In deeper water, faulting remained subdued through the Campanian and Maastrichtian (84–65 Ma). Reflector patterns are complex and result from minor relative sea-level changes, the broad shelf geometry and modest accommodation space.

From the end of the Cretaceous, the Latrobe Group and Taranaki's Kapuni coal measures and Moa Group marine facies are equivalent, even to deposition in a transgressive environment. Near the end of the Paleocene, minor volcanism is recorded across the flank of the Challenger Plateau synchronous with volcanism in onshore Gippsland.

Overall patterns are also similar between the two regions through the Neogene with the exception that Taranaki received voluminous clastic sediments and Gippsland was relatively starved of clastics.

Petroleum

Taranaki is New Zealand's only producing petroleum basin to date, while Gippsland has been Australia's most prolific petroleum region. Although the two basins were separated by at least 1,000 km at all times since formation until the Tasman Sea started spreading, both basins were at similar latitudes and environmental controls on deposition of coaly source rocks were likely to have been similar. Much of the petroleum in both Gippsland and Taranaki was sourced from coal measures of Cretaceous and Paleogene ages. In Taranaki, minor volumes were also generated from Paleocene black marine shale. Similarly, although coal is the major source rock in Gippsland, a marine contribution is detected in oils from the southeastern, oceanward part of the basin (Gorter, 2001).

Reservoir rocks in Gippsland are commonly of Paleocene age, although accumulations have also been discovered in Late Cretaceous sediments. Petroleum in Taranaki has been discovered mainly in Paleocene reservoirs of the Kapuni Group, but significant discoveries have also been made in fractured Oligocene limestone and in Neogene turbidite fans. The younger reservoirs are probably symptomatic of Neogene tectonism.

The results of the Astrolabe survey have interesting implications for both basins. They show that Cretaceous plays, particularly Rakopi Formation equivalents, are valid exploration targets and that marine source rocks should be considered in the deeper parts of both basins. The improved analogies between the basins may allow

conclusions to be drawn where knowledge of an aspect of one of the basins is poorly known and, while such circular arguments are not proof, such mutual support may be beneficial to exploration of both basins.

CONCLUSIONS

1. Before the Astrolabe survey, the Taranaki Basin was a relatively poor analogue for Gippsland as its oldest known sediments were much younger than Gippsland's. A relatively good comparison could be made between Late Cretaceous and Paleogene units of both basins, but the effects of the Neogene plate boundary are much more marked in Taranaki than in Gippsland. Taranaki received voluminous Neogene clastic sediments while Gippsland remained starved of them.
2. The Astrolabe survey provided evidence to suggest that Taranaki was probably initiated at about the same time as Gippsland and that correlatives of all of the main Gippsland sedimentary groups are likely to be present in the Taranaki Basin. The survey also revealed the presence of a large Late Cretaceous delta with extensive associated coal-bearing units, similar in age and tectonic setting to the Emperor and Golden Beach sub-groups and effectively linking Taranaki's Taniwha and Rakopi Formations to Gippsland equivalents.
3. Despite being separated by at least 1,000 km for their entire history, both basins were at the same latitude for much of their history, so source rocks in Taranaki should be at least as oil-prone as in Gippsland, assuming similar climatic controls on deposition and chemistry. Additionally, the location of Taranaki, at the head of a restricted marine trough, enhances the possibility that rich marine source rocks may be present. About 30% of the world's known oil reserves were generated from Cretaceous source rocks.

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